

Theme 3: Community Energy, Equity and Sustainability

Energy transition must be inclusive. Some communities face higher energy burdens, more outages, and greater climate risks. Utilities need tools to identify and support vulnerable populations. This theme focuses on equity, access, and community level sustainability.

Problem Statement 1:

Low-income households spend a higher percentage of income on energy, and many incentives are targeted at property owners rather than renters. Because renters pay the utility bills but do not control building upgrades, this misalignment can limit adoption of energy-efficiency measures. Utilities need simple tools to visualize where these gaps exist and identify communities facing higher energy burdens.

Possible solution(s):

- Energy Burden Index: Build an index combining income, building age, renter/owner mix, and estimated consumption.
- Energy Poverty Map: Design a map that highlights communities facing high energy costs or high renter populations where incentives may have lower uptake.
- Program Recommendation Tool: Design/propose a sample visualization showing targeted outreach, rebates or retrofit programs that could have greatest impact.

Datasets Needed:

Demographics, income, building age, energy consumption.

Public Dataset Sources (Examples):

- Statistics Canada Income Data: <https://www150.statcan.gc.ca/n1/en/type/data>
- Living Atlas – Demographics: <https://livingatlas.arcgis.com>
- Toronto Neighbourhood Profiles: <https://open.toronto.ca/>
- Peel Region Demographics: <https://opendata.peelregion.ca/>
- Spatial Access Measures (could be useful for transportation targeted solutions): <https://www150.statcan.gc.ca/n1/pub/27-26-0001/272600012023001-eng.htm>

- Canadian Index of Social Resilience and Canadian Index of Social Vulnerability: <https://www150.statcan.gc.ca/n1/en/catalogue/45200001>
- Enrichment data in ArcGIS Online (census and consumer-behaviour variables – included with student accounts): <https://doc.arcgis.com/en/esri-demographics/latest/esri-demographics/demographics.htm>

Problem Statement 2:

Climate impacts such as heatwaves and flooding disproportionately affect vulnerable populations. Many communities also rely on public facilities like libraries, schools, and community centers as warming or cooling spaces during extreme weather. Utilities and municipalities need simple ways to visualize where climate risks and limited access to safe shelter spaces overlap.

Possible solution(s):

- **Vulnerability Map:** Create a map that combines heat islands, outages, demographics, flood zones, and proximity to warming/cooling centers or community facilities.
- **Resilience Investment Tool:** Design a tool that ranks neighborhoods based on climate vulnerability.
- **Preparedness App Mockup:** Create a conceptual app that helps communities prepare for extreme weather.

Datasets Needed:

Heat Island maps, outage history, demographics, flood zones, public facility locations.

Public Dataset Sources (Examples):

- **Toronto Heat Vulnerability Index:** <https://open.toronto.ca/>
- **NRCan Flood & Heat Hazard Maps:** <https://open.canada.ca>
- **Living Atlas – Environmental Justice:** <https://livingatlas.arcgis.com>
- **Peel Region Climate Layers:** <https://opendata.peelregion.ca/>
- **Esri Canada Climate Change Hub:** <https://climate.esri.ca/>

Problem Statement 3:

Pollution sources and environmental hazards often cluster near marginalized communities. Utilities and municipalities need visibility into these patterns.

Possible solution(s):

- Pollution Exposure Map: Build a map that highlights communities exposed to pollution.
- Equity Planning Tool: Design a tool that identifies neighborhoods needing targeted sustainability programs, optionally including the presence of renewable energy installations to show where clean-energy investments may help offset environmental burdens.
- Environmental Justice Dashboard: Create a dashboard that visualizes pollution, demographics, and health indicators.

Datasets Needed:

Pollution data, demographics, land use, wind data, environmental layers.

Public Dataset Sources (Examples):

- Environment Canada Pollution Data: <https://pollution-waste.canada.ca/>
- Living Atlas – Environmental Justice Layers: <https://livingatlas.arcgis.com>
- Toronto Air Quality: <https://open.toronto.ca/>
- Ontario Environmental Registry: <https://ero.ontario.ca/>
- Canadian Index of Social Resilience and Canadian Index of Social Vulnerability: <https://www150.statcan.gc.ca/n1/en/catalogue/45200001>
- Enrichment data in ArcGIS Online (census and consumer-behaviour variables – included with student accounts): <https://doc.arcgis.com/en/esri-demographics/latest/esri-demographics/demographics.htm>